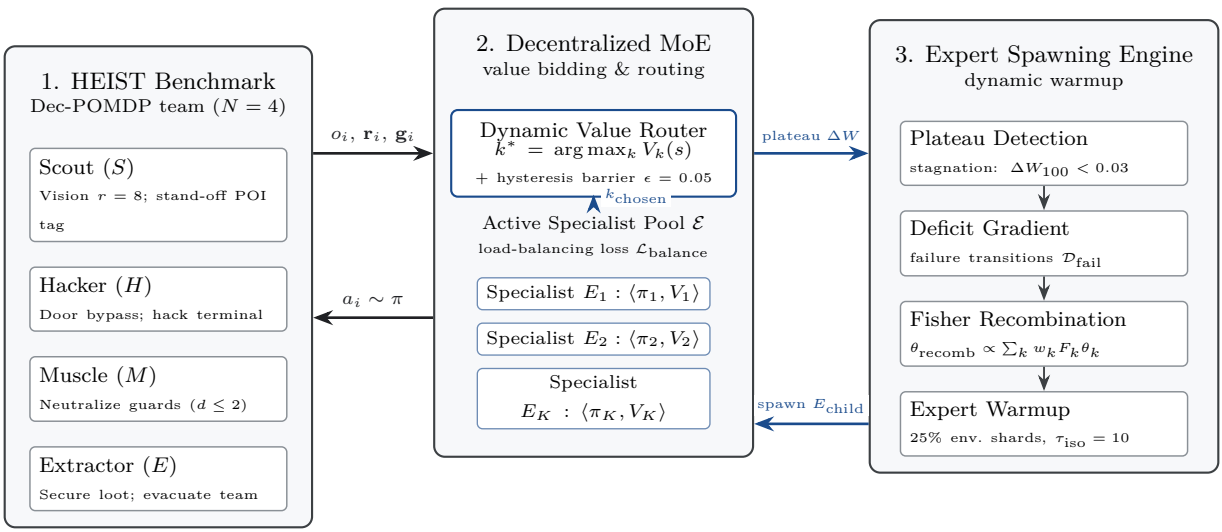
1. HEIST Benchmark
Dec-POMDP team ($N = 4$)
Scout (S)
Vision $r = 8$; stand-off POI tag
Hacker (H)
Door bypass; hack terminal
Muscle (M)
Neutralize guards ($d \leq 2$)
Extractor (E)
Secure loot; evacuate team
$o_i,\ \mathbf{r}_i,\ \mathbf{g}_i$
$a_i \sim \pi$
2. Decentralized MoE
value bidding & routing
Dynamic Value Router
$k^* = \arg\max_k V_k(s)$
+ hysteresis barrier $\epsilon = 0.05$
k_{chosen}
Active Specialist Pool $\mathcal{E}$
load-balancing loss $\mathcal{L}_{\text{balance}}$
Specialist $E_1 : \langle \pi_1, V_1 \rangle$
Specialist $E_2 : \langle \pi_2, V_2 \rangle$
Specialist
$E_K : \langle \pi_K, V_K \rangle$
plateau ΔW
spawn E_{child}
3. Expert Spawning Engine
dynamic warmup
Plateau Detection
stagnation: $\Delta W_{100} < 0.03$
Deficit Gradient
failure transitions $\mathcal{D}_{\text{fail}}$
Fisher Recombination
$\theta_{\text{recomb}} \propto \sum_k w_k F_k \theta_k$
Expert Warmup
25% env. shards, $\tau_{\text{iso}} = 10$